

Krishna Tripathi

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Education

Indian Institute of Information Technology, Nagpur
Bachelor of Technology in Computer Science and Engineering

December 2021 to May 2025
Nagpur, India

Technical Skills

Languages: C++, Go, Java, Python, JavaScript, TypeScript, SQL
Backend: Node.js, Express.js, FastAPI, REST APIs, Microservices
Frontend: React, Next.js **AI/LLM:** LangGraph, MCP, RAG, Langfuse
Databases: YugabyteDB, PostgreSQL, MongoDB, BigQuery, pgvector
Cloud/DevOps: AWS (Cognito, Lambda, S3), Google Cloud Run, Docker, Jenkins, GitHub Actions

Professional Experience

Yugabyte

Bengaluru, India

Software Engineer I

September 2025 to Present

- Engineered **Meko** knowledge base uploads across 3 services using React, Go, Python, and presigned S3 POST requests. Enforced 3 storage edge checks for file size, type, and content spoofing while offloading large transfers from API servers.
- Built **Meko's** retrieval observability and context promotion pipeline with top-k retrieval across memory and knowledge base sources. Replaced full conversation replay, reducing input token usage by **35%** for marketing and lead generation workflows.
- Delivered **AWS Cognito authentication across 3 identity providers:** Google, LinkedIn, and GitHub. Supported profile completion and identity linking while hardening invite, logout, and federated account flows.
- Contributed **AI/RAG integrations across Java, Python, and C++:** built LangChain4j embedding and chat-memory support plus a PgDistRagRetriever with C++ tablet-server changes for [pg_dist_rag](#). Added HNSW/hybrid search, metadata filters, integration tests, [docs](#), and a merged upstream [commit](#).
- Engineered **xCluster disaster recovery services in Go and UI**, optimizing DR node selection from **450ms to 120ms** and exposing C++ replication lag metrics through Prometheus and UI monitoring.
- Developed a **database migration validator for 2 health signals:** schema version and dirty state. Blocked invalid migrations before merge and surfaced actionable GitHub feedback.
- Improved **YugabyteDB's open-source developer ecosystem** through JavaScript driver logging, ORM load-balancing examples, and integration guidance.

Software Engineer Intern

September 2024 to August 2025

- Built the **end-to-end YB Voyager workflow in yugabyted UI across 4 migration stages**, adding status and progress analytics and rebuilding the [migration assessment](#) UX for clearer results. ([PR #2564](#) | [implementation](#))
- Built **growth analytics APIs and Go data workflows across 5 systems:** Scarf, GCS, BigQuery, MadKudu, and Salesforce, supporting centralized metrics, lead scoring, and CRM synchronization.
- Improved **yb-voyager developer tooling across 3 workflows:** added Homebrew install tracking, fixed Go/Docker setup in Jenkins helpers, and standardized build-script versioning; also fixed assessment HTML and moved stable-version testing to Jenkins. ([PR #2652](#) | [PR #2779](#))
- Improved **YugabyteDB developer onboarding** by adding [TypeORM smart-driver documentation](#) and repairing [Linux/RHEL setup links](#). ([PR #28250](#) | [PR #28418](#))

Zluri

Bengaluru, India

Software Engineer Intern

August 2024 to September 2024

- Resolved **13+ production bugs** in JavaScript/Node.js and MongoDB. Built QA dashboards and a testing framework, increasing coverage by **20%** and query performance by **30%**.

Projects

EternalAgni: Agnihotra Reminder App

July 2026 to Present

JavaScript, Capacitor, Android, PWA, Service Workers, Local Notifications

[SOURCE CODE](#) | [INTERNAL TESTING](#)

- Built a custom **NOAA timing engine** with ± 1 -second accuracy. Generated sunrise and sunset schedules for **92 days (~3 months)** locally for offline use.
- Delivered **reliable reminders across 2 platforms:** Android and PWA, with **2-to-60-minute** lead times, duplicate suppression, late delivery protection, client-side scheduling, GPS/IP fallbacks, service worker recovery, and automated APK/AAB builds.

YB Voyager Copilot: AI-Powered Database Migration Assistant

January 2025 to Present

LangGraph, FastAPI, Next.js, Python, MCP, Docker, YugabyteDB, RAG

[DOCKER HUB](#) | [ARCHITECTURE](#) | [DEMO](#)

- Architected a **3-agent LangGraph and MCP workflow** covering all 4 migration stages with grounded product documentation.
- Built a **3-tier RAG backend and containerized 5 services** with FastAPI, pgvector, YugabyteDB, and Docker; automated image publishing with GitHub Actions and Playwright PDF assessments with production sizing recommendations.

Achievements

- Received **Best Use of AI** recognition for YB Voyager Copilot at the YugabyteDB internal hackathon.
- Earned **Guardian status with a maximum rating of 2145** on LeetCode. Reached **4-star status with a 1944 rating** on CodeChef and **Specialist status with a maximum rating of 1484** on Codeforces.
- Secured **3rd place** in Algorithmia at Tantrafiesta 2023.